

Battery Charger With Solenoid Lock(+12V)

S_01_Revision_ History



File: S_01_REVISION_ HISTORY.kicad_sch

S_02_Block_Diagram



File: S_02_BLOCK_DIAGRAM.kicad_sch

S_03_Connectors



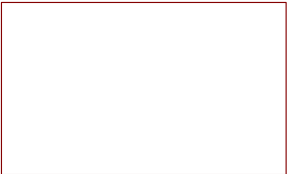
File: S_03_Connectors.kicad_sch

S_04_H-Bridge



File: S_04_H-Bridge.kicad_sch

S_05_Charger



File: S_05_Charger.kicad_sch

S_06_Microcontroller



File: S_06_NUVOTON_CONTROLLER.kicad_sch

S_07_Mechanical



File: S_07_MECHANICAL.kicad_sch

Project Name: SCH-BQBCHBD-2528-3A-01		
Size: A4	PCB Revision:	Schematic Revision: 3A
Date: 2025-07-08	Time:	Sheet: 1 Of 8
Drawn By: Anurag Raulji		
Reviewed By: Chirag Rana		
Approved By:Hardik Seth		
KiCad E.D.A. kicad 7.0.9		

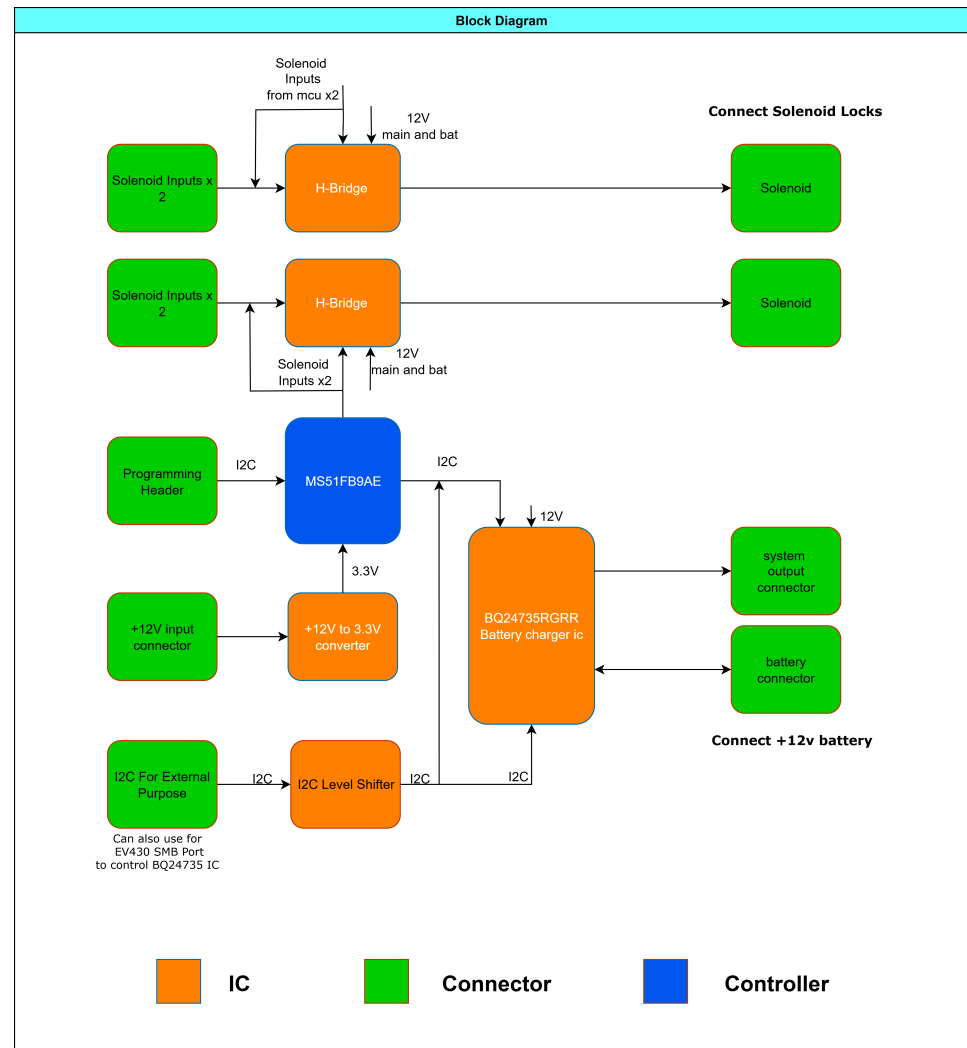


REVISION HISTORY

SR.NO.	DATE	REVISION	DESCRIPTION
1	17-jan-2022	1A-01	- Released initial version of schematic.
2	2-feb-2023	2A-01	- Added Nuvoton MS51FB9AE
3	8-july-2023	3A-01	- Added Solenoid Lock Logic with different mosfet then older version - Added Adapter voltage detection and battery voltage measurement - Separate Connector for System and Battery - Changed Filter capacitor footprint for controller

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Block Diagram



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Drawn By: Anurag Raulji

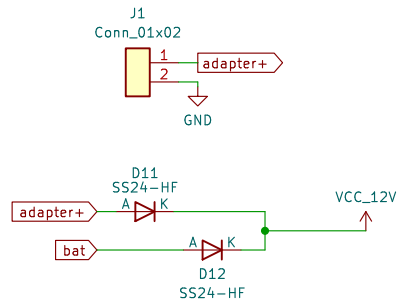
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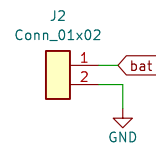
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Connectors

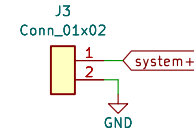
Input Supply



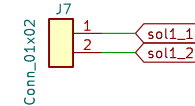
Battery



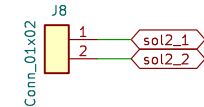
Output



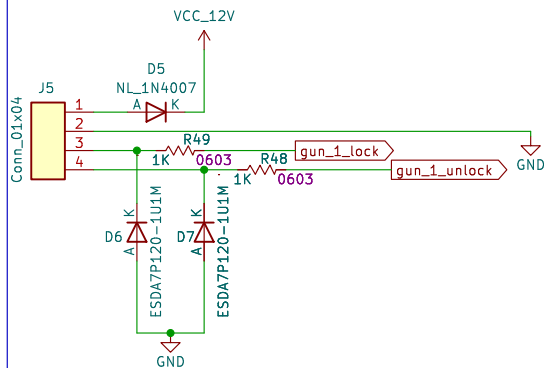
Solenoid-1



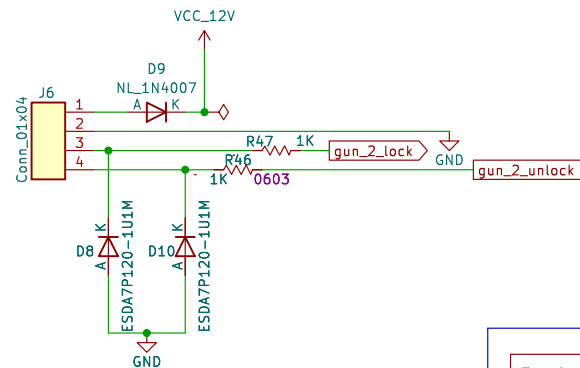
Solenoid-2



Input Solenoid-1



Input Solenoid-2



Project Name: SCH-BQBCHBD-2528-3A-01

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Drawn By: Anurag Raulji

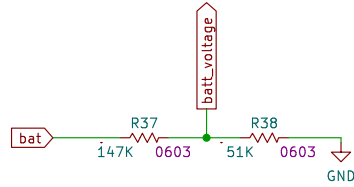
Reviewed By: Chirag Rana

Approved By: Hardik Seth

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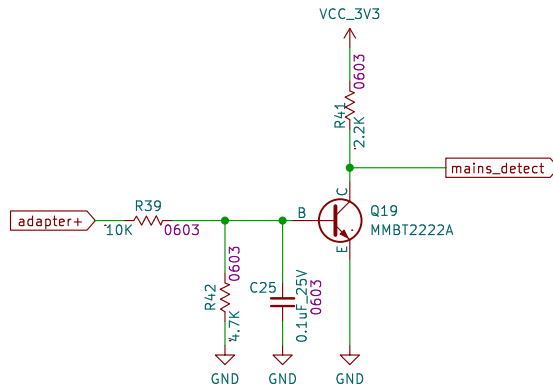
BACANCY
SYSTEMS PVT LTD

Battery Voltage Measurement

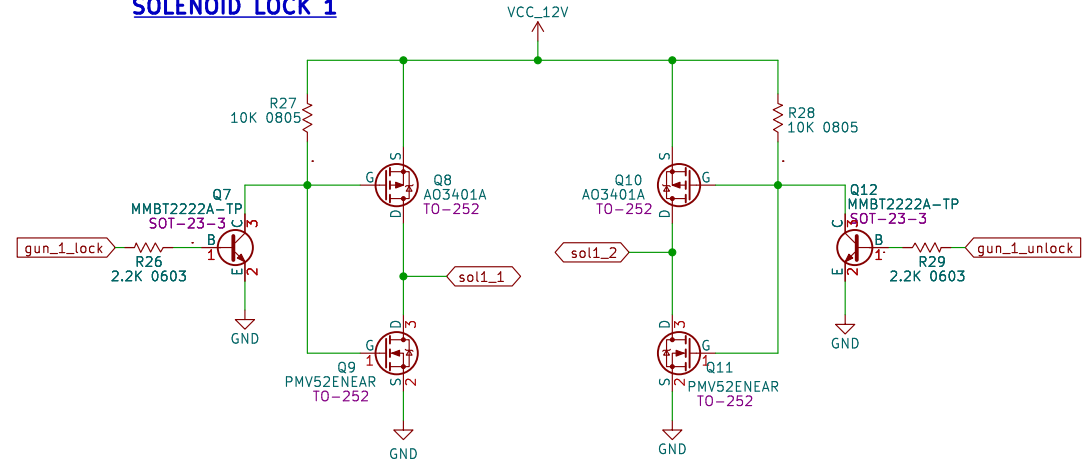


Input(V)	ADC(V)
9	2.318
9.5	2.447
10	2.576
10.5	2.705
11	2.833
11.5	2.962
12	3.091

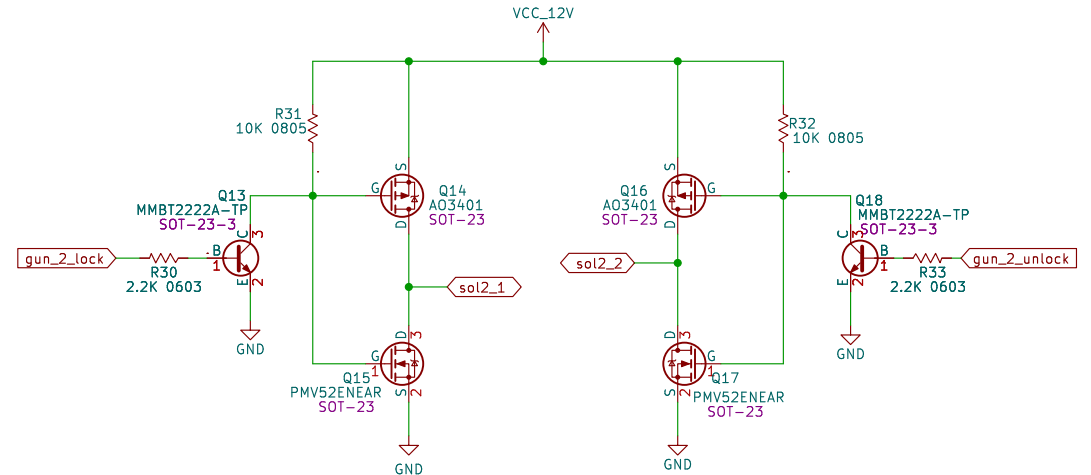
Mains Detection



SOLENOID LOCK 1



SOLENOID LOCK 2



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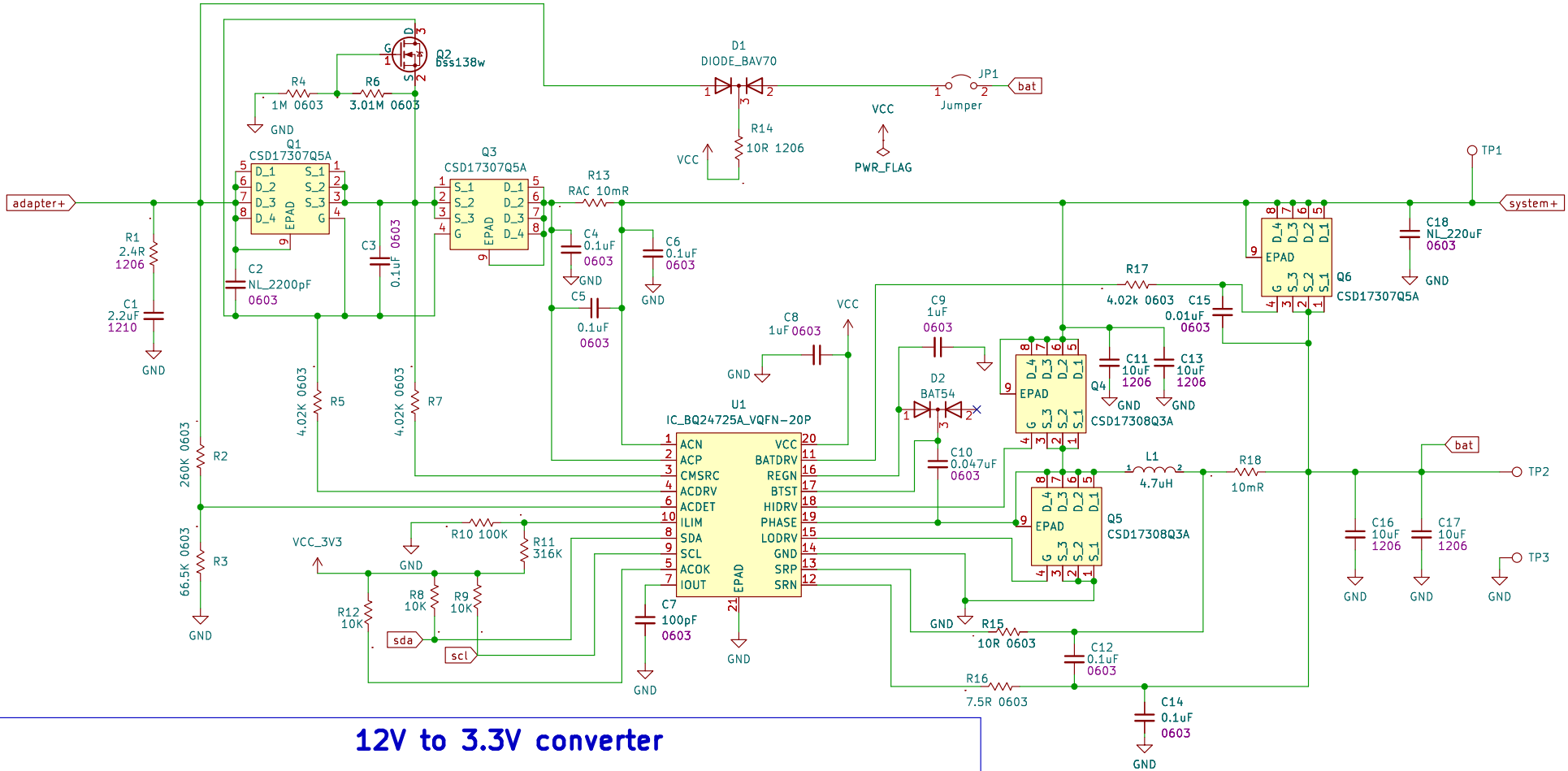
Drawn By: Anurag Raulji

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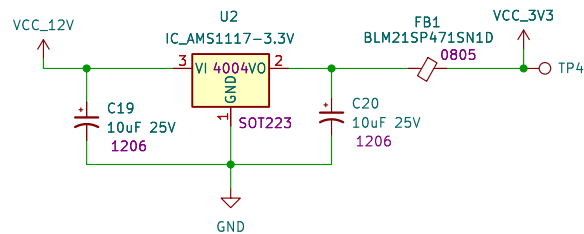
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Charger IC



12V to 3.3V converter



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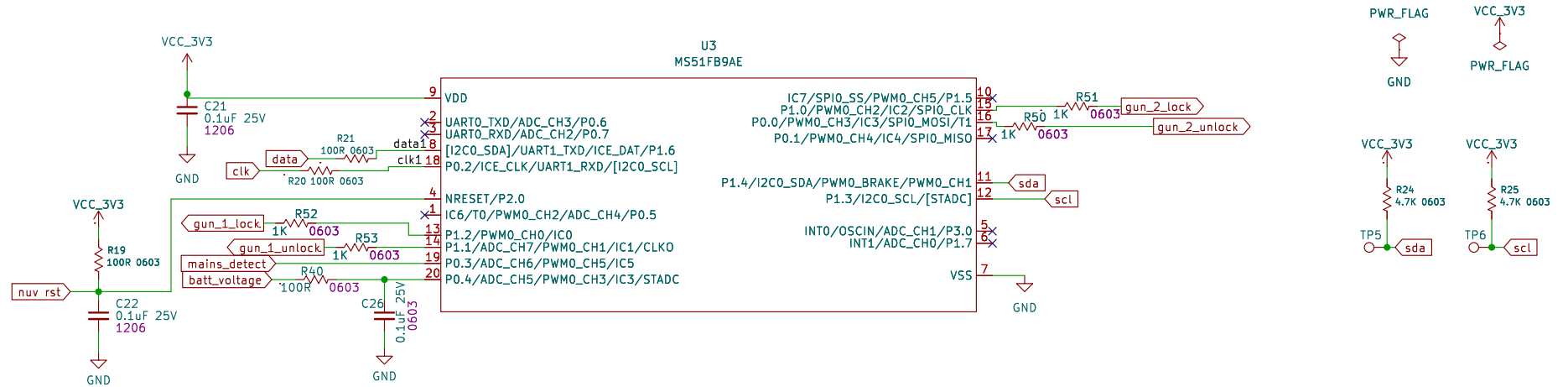
Reviewed By: Chirag Rana

Approved By: Hardik Seth

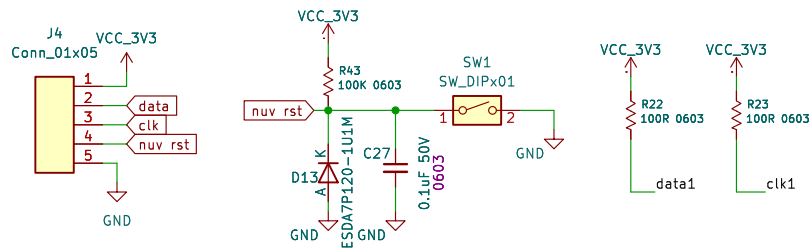
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Microcontroller



Programming Header



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Approved By: Hardik Seth

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Mechanical

MH1
MountingHole

MH3
MountingHole

FD1
Fiducial

FD2
Fiducial

MH2
MountingHole

MH4
MountingHole

FD3
Fiducial

FD4
Fiducial

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